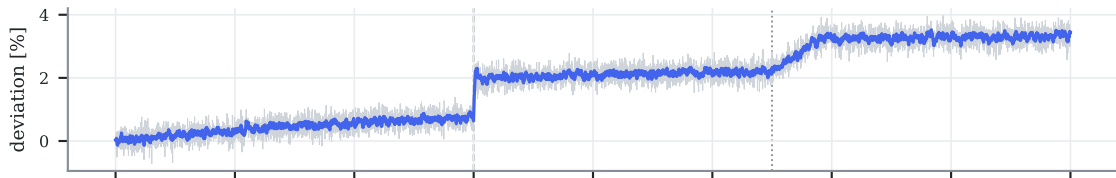
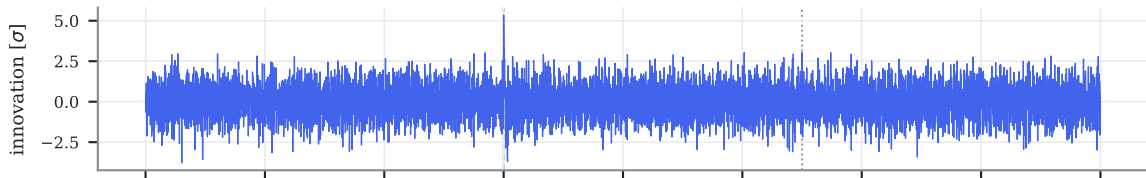


measured deviation (gray) and Holt level (blue); step (dashed) and ramp (dotted) onsets



Holt innovations: the chronic trend is gone; the step shouts, the slow ramp stays inside the noise



CUSUM on the innovations: fires on the step, blind to the ramp



dual-EWMA gap: opens by the fault magnitude on both events (threshold dashed)

